Mohammed Ishaaq

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Available for internship — Remote / Bengaluru — Immediate

Summary

Third-year Information Science & Engineering student who solves real problems using AI and software. Built a Government of India patented waste segregation system, reduced deep learning compute by 95%, and shipped systems demonstrated at national-level exhibitions and hackathons. Actively seeking a paid internship in AI, Machine Learning, or Software Development where I can contribute from day one.

Technical Skills

Programming	Python, C++, JavaScript, HTML/CSS
ML / AI	PyTorch, TensorFlow, Keras, Deep Learning, Computer Vision, Curriculum Learning
Data	NumPy, Pandas, Matplotlib, Seaborn
Tools	Git, GitHub, Google Colab, VS Code, Raspberry Pi

Projects

Staged Embarrassment Learning (SEL) — ishaaqdev/Staged-Embarrassment-Learning

- Engineered a deep learning training framework that solves wasted compute in resource-constrained environments – reduced operations by **95%** (33.5T → 0.34T FLOPs) while maintaining **84.7% accuracy on CIFAR-10**
- Implemented entropy-based dynamic filtering to skip redundant training samples, reducing unnecessary back-propagation across 99% of updates – designed for efficient deployment on low-resource hardware
- Reduced wall-clock training time by **42%** using a 5-stage curriculum learning pipeline on NVIDIA T4 GPU
- Built a custom real-time monitoring library (Python/JS) to visualize training efficiency, gradient sparsity, and loss dynamics
- Validated on the full CIFAR-10 benchmark (60,000 images); results logged and reproducible via public repository

AI Household Waste Segregator — ishaaqdev/IWM-Reverse-Vending-Machine

- Built a CNN-based computer vision model to solve real-world waste misclassification – resulted in a **Government of India Design Patent** (No. 482604-001)
- Implemented custom preprocessing and augmentation pipeline using TensorFlow/Keras; tested across varied real-world lighting conditions
- Optimized and deployed on Raspberry Pi for edge inference; demonstrated live at **Smart India Hackathon** – selected as Grand Finalist
- Prototype showcased at JVTM national engineering exhibition; validated by BGSCET Innovation Cell, securing **Rs. 1,00,000+** in funding

OceanOS – Ocean Pollution Response System — ishaaqdev/OceanOS

- Built a full-stack simulation to address real-time environmental monitoring – visualizes live IoT data (pH, TDS levels) in a 3D environment using Three.js and Python
- Engineered a closed-loop dashboard connecting pollution detection with automated response triggers
- Demonstrated at hackathon with simulated real-world sensor data; end-to-end pipeline functional from data ingestion to visual output

Recylo – Reverse Vending Machine — Hardware + Software

- Developed software layer and validation logic for an automated PET bottle recycling system
- Prototype physically built and demonstrated at JVTM technical exhibition; secured **Rs. 1,00,000+** in incubation funding

JobCraft – AI Career Roadmap System

- Prototyped a personalized learning path generator that analyses user skill gaps to produce step-by-step career roadmaps
- Integrated large language models to map technical skills to industry-standard job requirements

Innovation & Achievements

Design Patent — Government of India	Patent No. 482604-001, granted for an intelligent waste segregation device.
Grand Finalist — Smart India Hackathon	Selected among top national teams in India's premier government-run hackathon.
Rs. 1,00,000+ Incubation Funding	Awarded by BGSCET Innovation Cell for AI waste segregation and recycling systems.
Winner — Jnana Vijnana Tantrajnana Mela	National-level engineering innovation exhibition.
2nd Place — ZeroOne Hackathon	

Education

B.E. Information Science and Engineering — 2023 – Present
BGS College of Engineering and Technology, Bengaluru (VTU) — CGPA: 8.06

PUC – Science Stream
Falcon PU College — 84%

KSEEB Class 10
St Joseph's Indian High School — 74%

Problem Solving

Actively solving Data Structures and Algorithms on LeetCode using Python, covering arrays, strings, recursion, and dynamic programming.

Areas of Interest

Artificial Intelligence • Deep Learning • Computer Vision • Reinforcement Learning • Agentic AI Systems • Efficient ML for Edge Devices